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RESEARCH FOCUS

I am seeking a funded doctoral position in construction management. My core interest is distributed decision-making in construction project teams, built on ten years moving between materials research and organisational research, and informed by my MEng thesis finding that teams which spread decision-making capacity across levels were more resilient under compounding uncertainty than teams run by strong individual leaders. Around that spine sit genuine secondary interests: leadership and organisational behaviour, risk and crisis management, infrastructure delivery, procurement and supply-chain coordination, housing and project governance, and sustainable and circular construction implementation.

My methodological grounding is in qualitative and mixed-methods empirical research: semi-structured expert interviewing, survey design, manual thematic analysis, and multiple case study design. I am based in Germany and open to international relocation for the right funded position.

CORE COMPETENCIES

- **Research and Analysis:** Construction Management Research | Distributed Decision-Making | Organisational Behaviour and Leadership | Team Resilience Under Uncertainty | Mixed-Methods Empirical Research
- **Construction and Engineering:** Sustainable and Circular Construction | Construction Materials Research | Geotechnical Engineering | Infrastructure Delivery | Construction Economics
- **Delivery and Coordination:** Risk and Crisis Management | Procurement and Contract Administration | Multi-Stakeholder Coordination | Programme and Milestone Scheduling
- **Leadership and Development:** Industrial Supervision and Mentorship | Curriculum and Training Design | Cross-Cultural Team Coordination | Leadership Development

SKILLS

- **Qualitative Research Techniques:** semi-structured expert interviewing, reflexive thematic analysis (Braun and Clarke 2006), multiple case study design (Yin 2018), grounded and inductive coding approaches, purposive sampling, triangulation, research ethics approval processes.
- **Research Tools:** manual qualitative coding across two full coding passes; bilingual survey instrument design; Microsoft Excel, applied to survey data management and analysis.
- **Engineering and Technical Software:** Microsoft Project, AutoCAD, Revit Architecture, STAAD.Pro, Civil CADD (formal training completed); geotechnical laboratory testing including compressive strength, flexural strength, thermal conductivity, permeability, Proctor compaction, and CBR testing.
- **Writing and Documentation:** sole-authored thesis-length academic writing, structured literature review, survey instrument design, technical report writing, grant and research proposal writing (evidenced by a successful OMR 4,000 FURAP grant application).
- **Languages:** Bangla (native); English (professional working proficiency); Hindi (intermediate); German (A1).

EDUCATION

MEng International Project Management, Building, Real Estate and Infrastructure

HFT Stuttgart (Hochschule für Technik Stuttgart), Germany | 2021 to 2024

- Overall grade 1.9; thesis grade 1.7 (German scale, where 1.0 is the highest achievable grade).
- Thesis: Leading Through Crisis: Identifying and Cultivating Leadership Skills in Construction. Sole-authored.
- Mixed-methods design: structured literature review; bilingual workforce survey (350 practitioners approached, 92 responses); nine semi-structured interviews using a 32-question protocol; manual thematic analysis (Braun and Clarke 2006) across two coding passes.
- Key finding: teams that spread decision-making capacity across levels proved more resilient under compounding uncertainty than teams run by strong individual leaders.
- Supervised by Prof. Jürgen Marc Volm and Prof. Dr. Uta Bronner.
- Coursework: Sustainability and Green Building; Research Methods and Strategic Decision-Making; Leadership and Change Management; Construction Economics; Risk Management; Contract Law.

BEng (Hons) Civil Engineering

Caledonian College of Engineering, Oman (affiliated with Glasgow Caledonian University, Scotland) | 2014 to 2018

- Second Class Honours, Upper Division; CGPA 3.09 out of 4.0. Runner-up, Star Student Award.
- Thesis: A Comparative Study of Compaction Energy on Lime and GGBS Stabilised Soil.
- Sourced GGBS from Sohar Steel Industry and tested it as a partial lime substitute across four substitution ratios under standard and modified Proctor compaction.

- GGBS at 6 to 9 per cent substitution matched or beat lime's bearing capacity on its own, with the added benefit of diverting industrial by-product from landfill.

RESEARCH EXPERIENCE

Research Assistant and Data Analyst (part-time) | MCTspaceLab, Muscat, Oman | Fieldwork in Oman and Lisbon, Portugal | Jul 2024 to present

MCTspaceLab is an Omani research company working in collaboration with the Städtebau-Institut, Universität Stuttgart.

In Oman:

- Supported research on the livelihood strategies of low-income Bangladeshi migrant workers in Oman.
- Conducted exploratory interviews to inform the development of the research framework.
- Carried out in-depth interviews that were later used for life history mapping.
- Assisted with participant observation in workers' everyday environments.
- Collected visual and sensory data through photography and sound recordings to document the lived environment.
- Conducted public life studies by observing and mapping activities in markets and other public spaces, including pedestrian counts and activity mapping.
- Helped document and interpret on-the-ground research realities and contextual conditions to inform analysis.

In Lisbon, Portugal (from August 2026):

- Supporting research with second-time migrants who had relocated from Gulf countries to Lisbon.
- Conducting exploratory interviews to identify key themes and participant experiences.
- Carrying out semi-structured interviews with research participants.
- Undertaking participant observation to capture everyday practices and contextual dynamics.

Master's Thesis Researcher | HFT Stuttgart, Germany | 2023 to 2024

- Designed and ran the full research programme independently, from literature synthesis and survey design through ethics approval and recruitment.
- Ran a bilingual workforce survey (350 practitioners approached, 92 responses) and nine semi-structured interviews with project managers, a CEO and a senior site engineer, using a 32-question protocol developed for this study.
- Coded every transcript by hand using thematic analysis, across two full coding passes.
- Synthesised survey and interview findings into a sole-authored thesis, graded 1.7.

Team Member, FURAP Research Project | The Research Council of Oman | 2015 to 2016

- Six-member undergraduate team, competitive grant of OMR 4,000, investigating PET waste as a fine aggregate replacement in concrete.
- My contribution centred on materials testing across compressive strength, flexural strength, thermal conductivity, and permeability, results analysis, and research and academic writing for the project report.
- The team found that concrete with up to 32 per cent PET replacement still reached C35-grade structural performance. This was the first funded research project of my career.

PUBLICATIONS AND RESEARCH OUTPUTS

Conference

- Annisa, S. E. and Pranto, S. A. (2026). Conference abstract, Deliver Safe and Social Housing (DASH) Conference, Belgrade, Serbia, 28 to 29 October 2026. Accepted; second author and presenting author.

Preprints and manuscripts

- Six sole-authored preprints (2026), openly available on SSRN (abstract IDs 7242619, 7242178, 7242118 and 7241638) and Zenodo (DOIs 10.5281/zenodo.21893706 and 10.5281/zenodo.21892978). Full list on ORCID: orcid.org/0009-0002-4916-8181.
- Manuscript under peer review with a Springer Nature journal (2026).

Theses and research reports

- Leading Through Crisis: Identifying and Cultivating Leadership Skills in Construction. MEng thesis (unpublished), sole-authored, HFT Stuttgart, 2024.
- Performance Analysis of Eco-Concrete with PET as Fine Aggregate Replacement. Undergraduate team research report, The Research Council of Oman, FURAP Call 3, 2016. Six-member team; my contribution: materials testing, results analysis, and research and academic writing.
- A Comparative Study of Compaction Energy on Lime and GGBS Stabilised Soil. BEng thesis (unpublished), sole-authored, Caledonian College of Engineering, 2018.

INDEPENDENT RESEARCH INTERESTS

The following are ongoing, self-directed inquiries arising from my professional practice, not published or peer-reviewed work. I am developing each with the intention of pursuing formal research and publication once I have gathered sufficient data.

Client Decision-Making and Behavioural Preference Patterns in Residential Interior Design

Observing patterns in how clients across the ZAP portfolio make decisions, weigh trade-offs, and express aesthetic and budget preferences, with a view to what these patterns suggest at the level of the business as a whole.

Understanding Client Psychology in Bespoke Residential Interior Delivery

Exploring how an individual client's motivations, priorities, and anxieties unfold over the course of a single bespoke project, informing how I manage that relationship directly.

Direct-Access Market Structures: Understanding Wholesaler-Retailer-Client Relationships in Bangladesh

Examining how the Bangladeshi construction materials market's unusual pattern of direct client access to wholesalers reshapes the traditional roles of retailers and intermediaries, and what that means for how a construction business should be positioned commercially.

TEACHING EXPERIENCE

Industrial Supervisor, University Engineering Placements | Muscat Laboratory Company, Oman | 2018 to 2020

- Supervised more than 500 undergraduate engineering students from five Omani universities across mandatory eight-week placements.
- Walked students through materials testing protocols, geotechnical investigation, and laboratory discipline, often for the first time in their working lives.
- Taught technical report writing and research proposal structure alongside the hands-on supervision.
- Gave individual written feedback on every student's technical work each placement cycle.

Invited Guest Lecturer | Multiple institutions, Oman | 2018 to 2021

- Taught Soil Mechanics, Building Evaluation, Material Testing, Sustainable Construction, and Academic Writing at Caledonian College of Engineering.
- Covered Construction Principles and Materials Engineering at Sultan Qaboos University.
- Delivered Material Testing and Building Evaluation sessions at the Ministry of Defence Engineering Department.

Private University Tutor | Oman, Bangladesh, and online | 2016 to 2025

- Provided paid academic support to roughly 325 to 375 university students over the period.
- Covered soil mechanics, construction technology, materials testing, presentation skills, and research methodology.
- Adjusted my approach for students coming from very different academic backgrounds and confidence levels.

Student Assistant, Programme Coordination | IUSD, Universität Stuttgart, Germany | 2022 to 2024

- Coordinated 17 academic events across 2.5 years while completing the MEng alongside the role.
- Managed intake communication each year for 30 to 40 international students from more than ten countries.
- Handled cross-cultural student communication in one of the university's most internationally diverse programmes.

TEACHING PHILOSOPHY

I do not teach by standing at the front of a room and talking. I bring students into the problem so they have to think, question, and argue their way to an answer, the same way I had to when I was the one on site with a decision to make and no clean textbook answer available. Real project examples do most of the work here. When a concept has actually failed somewhere, on a real project, students remember it in a way that a slide never manages.

In placement supervision I have watched a hesitant first-time report writer turn into someone who can present findings confidently to a laboratory client, inside a single eight-week placement. That shift, watching someone's confidence catch up with their competence, is what I am chasing in every teaching setting I take on. I treat whatever practical experience a student already has as something to build from, not something to correct with theory. And I write detailed feedback on actual technical work rather than handing back a single grade, because a number tells a student almost nothing about what to do differently next time.

TEACHING SUBJECTS

- **Core Teaching Areas:** Construction Management | Project Management | Soil Mechanics | Materials Testing and Technology | Geotechnical Investigation | Technical Report Writing | Research Methods | Sustainable Construction.
- **Also Comfortable Teaching:** Construction Economics | Risk Management | Procurement and Contract Administration | Site Management | Leadership and Organisational Behaviour | Built Environment Delivery | Civil Engineering Technology | Business Analysis Fundamentals.

PROFESSIONAL EXPERIENCE

Founder and Operations Lead | ZAP Comfortable Homes, Bangladesh | 2020 to present

- Started this business from nothing and carry sole responsibility for procurement, contractor management, quality, risk, and every client relationship on the books.
- Delivered more than 50 projects, running up to ten concurrent active sites at the busiest points.

- Negotiated 17 supplier contracts, bringing procurement cost down 25 to 30 per cent through direct commercial negotiation rather than accepting first quotes.
- Keep structured risk registers running across the live portfolio, so every decision point under time or resource pressure has a paper trail.
- Brought a flagship residential project in 60 days ahead of programme by resequencing trades early rather than reacting to delay after it happened.
- Built and run internal training for a 32-member team spanning finance, design, procurement, site engineering, and skilled trades.

Student Assistant | IUSD, Universität Stuttgart, Germany | Feb 2022 to Aug 2024

- Coordinated 17 international academic events across 2.5 years, including thesis defences, workshops, and international field trips, for audiences of up to 175 attendees, on time and within budget.
- Owned end-to-end international student intake for 30 to 40 students per cycle across more than ten nationalities, from application processing through interview panel coordination.

Civil Engineer and Marketing Coordinator | Muscat Laboratory Company, Oman | 2018 to 2020

- Delivered more than 250 geotechnical investigation reports on a consistent two-week turnaround, regardless of how the workload stacked up that month.
- Carried more than 110 projects through their full lifecycle, from initial scoping through to client delivery.
- Led geotechnical investigation on the Muscat International Airport Expansion, a genuinely high-pressure, multi-stakeholder infrastructure programme.
- Led geotechnical investigation on the Al Amerat Road Expansion and briefed Ministry of Transport stakeholders on progress every week.
- Completed more than 70 structural building evaluations for commercial and residential buildings, meeting full regulatory compliance with no findings on re-audit.
- Supervised teams of up to 15 engineers and technicians across concurrent site and laboratory work.

Co-Founder | Unique Electronics LLC, Muscat, Oman | Jan 2016 to Jun 2018

- Co-founded a small retail electronics initiative at age 20 with a business partner, managing procurement, inventory, and daily trading floor operations from an initial OMR 5,700 investment through to a trade-sale exit in 2018.
- Negotiated inventory contracts with regional suppliers and handled customer service and after-sales support.

Earlier Engineering Roles and Training | Oman and Egypt | 2014 to 2017

- Civil Engineering Training, Al Watanyiah United Engineering and Contracting, Oman (Feb to Apr 2017): hands-on civil engineering training across active sites covering site operations, technical documentation, and field engineering.
- Engineering and Business Analyst Intern, Al Mazaya Trading and Contracting, Oman (2014 to 2017): supported multiple construction projects through requirements analysis, stakeholder coordination, and project documentation.
- Civil Department Trainee, Modern Light Trading and Contracting, Oman (Jan to Mar 2016): cost estimation, procurement, construction methodology, and on-site operations exposure.
- Engineering Trainee, Technibuild Contracting LLC, Oman (Jul 2015): structural inspection, materials testing, risk analysis, and quality assurance.
- Engineering Intern, Arabisk Construction Company, Cairo, Egypt (Jan to Feb 2015): supported construction projects through data analysis, process mapping, and site management.

SELECTED PROJECTS

Muscat International Airport Expansion, Oman

Led geotechnical investigation across a high-pressure, multi-stakeholder infrastructure programme, coordinating site investigation and materials testing against a demanding delivery schedule.

Al Amerat Road Expansion, Oman

Delivered geotechnical investigation with weekly Ministry of Transport briefings, maintaining technical credibility and programme reliability throughout the engagement.

ZAP Flagship Residential Project, Bangladesh

Delivered a full residential construction project 60 days ahead of programme through proactive resequencing, early risk identification, and coordinated multi-trade delivery.

AWARDS AND RECOGNITION

- Research grant recipient, The Research Council of Oman, FURAP Call 3 (2015 to 2016). OMR 4,000 competitive grant to an undergraduate team.
- Winner, British Council Public Speaking Competition, Sultanate of Oman.
- Nominated Delegate, People to People World Leadership Forum, Washington, D.C.
- Runner-up, Star Student Award, Caledonian College of Engineering.

- Former President, AIESEC Oman, where I led a team of 25 through strategic planning and leadership development.

PROFESSIONAL DEVELOPMENT AND MEMBERSHIPS

- Certified Business Analyst Professional (CBAP), Qatar Advanced Technology and Research Centre.
- Formal technical software training completed at CADD Centre, Oman (see Skills for software covered).
- Former Member, Chartered Institute of Building (CIOB), United Kingdom.

SELECTED OPEN RESEARCH QUESTIONS

- How does distributed decision-making capacity develop and sustain itself in teams facing circular renovation and design-for-disassembly requirements?
- What supervisory conditions let information and permission to act reach every level of a team, rather than staying concentrated in leadership?
- How do industrial by-products like GGBS and PET move from validated lab findings to default specification on live projects, and what actually stops them?

REFEREES

Four academic and professional referees, including both MEng thesis supervisors, are available on request.